



COMPETENCY STANDARD

FOR

DFT AND VERIFICATION ENGINEERING

PROGRAM FOR ASIC DESIGNERS

(SEMICONDUCTOR / VLSI SECTOR)

09 August 2026

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for DFT and Verification Engineering Program for ASIC Designers serve as a foundational document for the development of curricula, teaching materials, and assessment tools specifically tailored for the Semiconductor/ VLSI sector. This document ensures that training activities align with industry requirements, enabling individuals who successfully meet the established standards through assessment to become qualified professionals in relevant roles within the sector.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical for DFT and Verification Engineering Program for ASIC Designers. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:
Generic Competencies (08 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-DVE-01-G	Apply OHS, ESD and Safe Workplace Practices	1. Identify OHS and ESD requirements 2. Practice safe workstation and EDA lab procedures 3. Follow emergency response procedures	04
SICIP-SCD-DVE-02-G	Apply Workplace Communication and Professional Ethics	1. Interpret workplace communication and etiquette 2. Prepare technical documents 3. Participate in design review and presentation	04
Total Hour			08 Hrs.

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-DVE-01-S	Interpret Semiconductor, ASIC, Digital Logic and DFT Fundamentals	1. Describe semiconductor and ASIC flow. 2. Explain digital logic and CMOS fundamentals. 3. Identify DFT concepts and terminology.	20
Total Hour			20 Hrs.

Occupation Specific Competencies (212 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SCD-DVE-01-O	Apply Linux, RTL and Verification Fundamentals	1. Operate Linux/UNIX environment. 2. Apply shell and TCL scripting. 3. Set up EDA environment. 4. Write Verilog RTL modules. 5. Design FSM and DFT-registers. 6. Create and simulate RTL testbench.	32
SICIP-SCD-DVE-02-O	Develop SystemVerilog Testbench and Verification Environment	1. Apply SystemVerilog data types. 2. Apply interfaces and testbench structure. 3. Apply object-oriented programming. 4. Develop testbench architecture. 5. Execute simulations and analyze waveforms. 6. Debug verification failures.	40
SICIP-SCD-DVE-03-O	Perform Coverage-Driven and UVM Verification	1. Write assertions. 2. Implement functional coverage. 3. Apply coverage-driven verification. 4. Apply UVM methodology. 5. Develop UVM components and sequences. 6. Implement checking and debugging.	44
SICIP-SCD-DVE-04-O	Apply Advanced ASIC Verification Techniques	1. Interpret timing and constraints. 2. Perform CDC/RDC verification. 3. Implement low-power verification concepts. 4. Execute lint, formal and equivalence checking.	24
SICIP-SCD-DVE-05-O	Implement Design for Testability (DFT)	1. Explain scan architecture. 2. Perform scan insertion setup.	36

		3. Debug scan DRC and violations. 4. Apply ATPG and fault model concepts. 5. Generate and analyze ATPG coverage. 6. Apply compression and timing-aware testing.	
SICIP-SCD-DVE-06-O	Apply Advanced Test and Silicon Validation Techniques	1. Explain boundary scan and JTAG. 2. Implement MBIST and LBIST concepts. 3. Interpret OPCG/OCC concepts. 4. Perform post-silicon diagnosis.	16
SICIP-SCD-DVE-07-O	Complete ASIC Verification and DFT Project	1. Plan project and prepare architecture. 2. Execute verification and DFT implementation. 3. Validate verification results. 4. Prepare project documentation. 5. Present project outcomes.	20
Total Hour			212 Hrs.

COMPETENCY STANDARD: DFT AND VERIFICATION ENGINEERING PROGRAM FOR ASIC DESIGNERS

The Generic Competencies

Unit of Competency: APPLY OHS, ESD AND SAFE WORKPLACE PRACTICES	Nominal Duration: 04 hrs.	Unit Code: SICIP-SCD-DVE-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply OHS, ESD and Safe Workplace Practices. It specifically includes the tasks of identifying OHS and ESD requirements, practicing safe workstation and EDA lab procedures and following emergency response procedures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS and ESD requirements	1.1 <u>Workplace OHS and ESD requirements</u> are identified according to organizational procedures. 1.2 <u>Hazards</u> associated with semiconductor and EDA laboratory environments are recognized. 1.3 <u>Personal protective equipment (PPE) and ESD protection devices</u> are identified and used appropriately. 1.4 ESD control measures are applied to protect electronic components and equipment. 1.5 <u>Workplace safety signs, symbols, and instructions</u> are interpreted correctly.
2. Practice safe workstation and EDA lab procedures	2.1 <u>Workstations</u> are prepared according to OHS and ESD requirements. 2.2 <u>EDA tools, computers, and laboratory equipment</u> are operated following safe work procedures. 2.3 <u>Cables, electrical equipment, and work areas</u> are maintained to prevent workplace hazards. 2.4 Good housekeeping practices are maintained to ensure a safe working environment. 2.5 Workplace incidents, hazards, and unsafe conditions are identified and reported promptly.
3. Follow emergency response procedures	3.1 <u>Emergency procedures</u> are identified according to workplace requirements. 3.2 Emergency exits, assembly points, and safety equipment are located and accessed correctly. 3.3 Appropriate actions are taken during electrical, fire, or other workplace emergencies.

	3.4 Incidents and accidents are reported according to organizational procedures. 3.5 <u>Emergency response activities</u> are carried out in accordance with workplace safety requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Workplace OHS and ESD requirements	1.1 OHS Policies 1.2 ESD Procedures 1.3 Workplace Rules 1.4 Safe Work Practices 1.5 Emergency Procedures 1.6 Risk Assessment
2. Hazards	2.1 Electrostatic Discharge (ESD) 2.2 Electrical Hazards 2.3 Fire Hazards 2.4 Chemical Hazards 2.5 Trip and Slip Hazards 2.6 Equipment Hazards
3. Personal protective equipment (PPE) and ESD protection devices	3.1 ESD Wrist Strap 3.2 ESD Shoes 3.3 ESD Mat 3.4 Safety Glasses 3.5 Protective Gloves 3.6 Lab Coat
4. Workplace safety signs, symbols, and instructions	4.1 Warning Signs 4.2 Mandatory Signs 4.3 Prohibition Signs 4.4 Emergency Exit Signs 4.5 Fire Safety Signs 4.6 Hazard Symbols
5. Workstations	5.1 ESD Workstation 5.2 Computer Workstation 5.3 Design Workbench 5.4 Laboratory Bench 5.5 Test Workstation 5.6 Simulation Workstation
6. EDA tools, computers, and laboratory equipment	6.1 EDA Software 6.2 Workstation Computer 6.3 Server 6.4 Oscilloscope 6.5 Digital Multimeter 6.6 Power Supply

7. Cables, electrical equipment, and work areas	7.1 Power Cables 7.2 Network Cables 7.3 Extension Cords 7.4 Electrical Outlets 7.5 Work Benches 7.6 Cable Management
8. Emergency procedures	8.1 Fire Emergency 8.2 Electrical Emergency 8.3 Medical Emergency 8.4 Evacuation Procedure 8.5 First Aid Procedure 8.6 Incident Reporting Procedure
9. Emergency response activities	9.1 Evacuation 9.2 First Aid 9.3 Firefighting 9.4 Hazard Isolation 9.5 Emergency Notification 9.6 Incident Documentation

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Principles of Occupational Health and Safety (OHS) and Electrostatic Discharge (ESD) protection. 1.2 Common hazards in semiconductor fabrication, EDA laboratories, and electronic work environments. 1.3 Proper use of Personal Protective Equipment (PPE) and ESD protection devices. 1.4 ESD control methods and safe handling of electronic components. 1.5 Safe operation of computers, EDA tools, electrical equipment, and laboratory workstations. 1.6 Workplace housekeeping practices and electrical safety requirements. 1.7 Emergency procedures, including fire safety, electrical hazards, evacuation routes. 1.8 Interpretation of workplace safety signs, symbols, and organizational safety policies and procedures.
2. Underpinning Skills	2.1 Applying workplace OHS and ESD requirements in semiconductor and EDA laboratory environments 2.2 Interpreting workplace safety signs, symbols, and safety instructions 2.3 Preparing and maintaining safe workstations and EDA laboratory environments 2.4 Operating EDA tools, computers, and laboratory equipment safely 2.5 Applying good housekeeping practices to maintain a hazard-free workplace

	2.6 Following emergency response procedures for electrical, fire, and workplace emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning/operational manual 4.3 PPEs 4.4 Firefighting equipment 4.5 Emergency response manual 4.6 Emergency Systems (Fire alarms, emergency exits, etc.) 4.7 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified OHS and ESD requirements 1.2 practiced safe workstation and EDA lab procedures 1.3 followed emergency response procedures
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY WORKPLACE COMMUNICATION AND PROFESSIONAL ETHICS	Nominal Duration: 04 hrs.	Unit Code: SICIP-SCD-DVE-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply workplace communication and professional ethics. It specifically includes the tasks of interpreting workplace communication and etiquette, preparing technical documents and participating in design review and presentation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 <u>Workplace communication methods</u> and organizational protocols are identified. 1.2 <u>Professional etiquette</u> is demonstrated during workplace interactions. 1.3 Technical information and work instructions are interpreted accurately. 1.4 <u>Verbal and written communication</u> is conducted clearly and professionally. 1.5 Team communication and collaboration are maintained according to workplace requirements.
2. Prepare technical documents	2.1 <u>Technical information</u> is collected and organized according to project requirements. 2.2 Design notes, reports, and technical documents are prepared using standard formats. 2.3 <u>Technical terminology and documentation standards</u> are applied correctly. 2.4 Documents are reviewed for accuracy, completeness, and consistency. 2.5 Final documents are submitted according to organizational procedures.
3. Participate in design review and presentation	3.1 <u>Design data and supporting materials</u> are prepared for review meetings. 3.2 Design concepts and technical information are presented clearly to stakeholders. 3.3 Feedback and review comments are recorded and analyzed. 3.4 <u>Design modifications</u> are incorporated based on review outcomes. 3.5 Professional communication and teamwork are demonstrated throughout the review and presentation process.

Range of Variables

Variable	Range May Include but not limited to:
1. Workplace communication methods	1.1 Face-to-Face Communication 1.2 Email 1.3 Phone Calls 1.4 Video Conferencing 1.5 Instant Messaging
2. Professional etiquette	2.1 Respectful Behaviour 2.2 Professional Language 2.3 Punctuality 2.4 Active Listening 2.5 Appropriate Dress Code 2.6 Workplace Ethics
3. Verbal and written communication	3.1 Meetings 3.2 Emails 3.3 Reports 3.4 Presentations 3.5 Memos 3.6 Technical Documentation
4. Technical information	4.1 Design Specifications 4.2 Project Requirements 4.3 Technical Drawings 4.4 Test Results 4.5 Design Data 4.6 Reference Documents
5. Technical terminology and documentation standards	5.1 Technical Terms 5.2 Industry Standards 5.3 Naming Conventions 5.4 Document Templates 5.5 File Formats 5.6 Documentation Guidelines
6. Design data and supporting materials	6.1 Design Drawings 6.2 Schematics 6.3 Presentation Slides 6.4 Technical Reports 6.5 Simulation Results 6.6 Design Specifications
7. Design modifications	7.1 Design Updates 7.2 Specification Changes 7.3 Layout Revisions 7.4 Documentation Updates 7.5 Error Corrections 7.6 Improvement Actions

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Principles of workplace communication, professional etiquette, and organizational communication protocols. 1.2 Methods for interpreting technical information, work instructions, and engineering documentation. 1.3 Standards and formats for preparing technical reports, design notes, and project documentation. 1.4 Use of technical terminology, documentation conventions, and version control practices. 1.5 Principles of effective presentation, technical discussion, and design review processes. 1.6 Methods for receiving, incorporating technical feedback into design improvements. 1.7 Teamwork, collaboration, and professional communication practices in engineering.
2. Underpinning Skills	2.1 Applying professional workplace communication and organizational etiquette 2.2 Collaborating with team members to support design and project activities 2.3 Preparing technical documents, reports, and design notes using standard formats 2.4 Applying technical terminology and documentation standards correctly 2.5 Reviewing technical documents for accuracy, completeness, and consistency 2.6 Presenting design concepts and technical information clearly during review meetings 2.7 Incorporating feedback to improve design quality and project outcomes
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Work books 4.3 Learning manuals 4.4 Technical documents

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted workplace communication and etiquette 1.2 prepared technical documents 1.3 participated in design review and presentation
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

Sector Specific Competencies

Unit of Competency: INTERPRET SEMICONDUCTOR, ASIC, DIGITAL LOGIC AND DFT FUNDAMENTALS	Nominal Duration: 20 hrs.	Unit Code: SICIP-SCD-DVE-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret Semiconductor, ASIC, Digital Logic and DFT Fundamentals. It specifically includes the tasks of describing semiconductor and ASIC flow, explaining digital logic and CMOS fundamentals and identifying DFT concepts and terminology.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Describe semiconductor and ASIC flow.	1.1 <u>Semiconductor industry structure</u> and ASIC applications are identified. 1.2 Semiconductor manufacturing, design, packaging, and testing processes are described. 1.3 <u>Front-end and back-end ASIC design flow</u> are interpreted. 1.4 Major design stages, deliverables, and EDA tools used in the ASIC flow are identified. 1.5 ASIC design documentation and project flow are interpreted according to project requirements.
2. Explain digital logic and CMOS fundamentals.	2.1 Boolean logic, logic gates, and combinational and sequential circuits are explained. 2.2 <u>CMOS technology and MOSFET operating principles</u> are described. 2.3 Timing concepts including clock, setup time, hold time, and propagation delay are interpreted. 2.4 <u>Basic digital circuits</u> are analyzed according to design requirements. 2.5 Digital logic and CMOS concepts are related to ASIC design and verification.
3. Identify DFT concepts and terminology.	3.1 <u>DFT objectives and benefits</u> are identified according to ASIC design requirements. 3.2 <u>Common DFT terminology</u> , scan architecture, and test structures are explained. 3.3 Fault models, test coverage, and ATPG concepts are identified. 3.4 DFT design flow and verification activities are interpreted.

	3.5 DFT reports are interpreted according to project requirements.
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Range of Variables

Variable	Range May include but not limited to:
1. Semiconductor industry structure	1.1 Semiconductor Industry Sectors 1.2 Consumer Electronics 1.3 Automotive Electronics 1.4 Telecommunications 1.5 Artificial Intelligence (AI) Hardware
2 Front-end and back-end ASIC design flow	2.1 RTL Design 2.2 Functional Verification 2.3 Logic Synthesis 2.4 Physical Design 2.5 Static Timing Analysis (STA) 2.6 Signoff and Tape-Out
3 CMOS technology and MOSFET operating principles	3.1 CMOS Technology 3.2 PMOS Transistor 3.3 NMOS Transistor 3.4 Cutoff Region 3.5 Triode (Linear) Region 3.6 Saturation Region
4 Basic digital circuits	4.1 Multiplexer (MUX) 4.2 Decoder 4.3 Encoder 4.4 Comparator 4.5 Adder 4.6 Counter
5 DFT objectives and benefits	5.1 Testability Improvement 5.2 Manufacturing Test Support 5.3 Fault Detection 5.4 Test Cost Reduction 5.5 Product Quality Improvement 5.6 Yield Enhancement
6 Common DFT terminology	6.1 Scan Chain 6.2 Scan Flip-Flop 6.3 Scan Enable 6.4 Boundary Scan (JTAG) 6.5 Test Access Port (TAP) 6.6 Built-In Self-Test (BIST)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Semiconductor industry, ASIC applications, and complete ASIC design flow. 1.2 Front-end and back-end design stages, EDA tools, and project documentation. 1.3 Digital logic fundamentals and sequential circuits. 1.4 CMOS technology, MOSFET operation, and digital circuit principles. 1.5 Timing concepts and propagation delay. 1.6 Fundamentals of Design for Testability (DFT), scan architecture, and test structures. 1.7 Interpretation of DFT reports and their role in ASIC quality and testing.
2. Underpinning Skills	2.1 Understanding semiconductor industry structure and the ASIC design flow 2.2 Interpreting front-end and back-end design processes, EDA tools, and project deliverables 2.3 Applying digital logic fundamentals and sequential circuits 2.4 Interpreting timing concepts such as clock, setup time, hold time, and propagation delay 2.5 Relating digital logic and CMOS principles to ASIC design and verification 2.6 Applying knowledge of scan architecture, fault models, test coverage, and ATPG concepts 2.7 Interpreting DFT design flow, verification activities, and DFT reports for ASIC projects.
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 described semiconductor and ASIC flow. 1.2 explained digital logic and CMOS fundamentals. 1.3 identified DFT concepts and terminology.
2 Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3 Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation-specific Competencies

Unit of Competency: DESIGN, LAYOUT AND VERIFY CMOS STANDARD CELLS	Nominal Duration: 32 hrs.	Unit Code: SICIP-SCD-DVE-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to Apply Linux, RTL and Verification Fundamentals. It specifically includes the tasks of operating Linux/UNIX environment, applying shell and TCL scripting, setting up EDA environment, writing Verilog RTL modules, designing FSM and DFT- registers, and creating and simulate RTL testbench.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Operate Linux/UNIX environment.	1.1 <u>Linux/UNIX environment</u> is accessed according to project requirements. 1.2 Linux commands are used to manage files, directories, and processes. 1.3 File permissions and environment variables are configured as required. 1.4 <u>Project files</u> are organized following workplace standards. 1.5 Linux environment is verified before executing EDA tools.
2. Apply shell and TCL scripting.	2.1 <u>Shell and TCL scripts</u> are developed according to task requirements. 2.2 Variables, procedures, and control statements are applied in scripts. 2.3 Scripts are executed to automate <u>design and verification tasks</u> . 2.4 Script execution results are verified and errors are corrected. 2.5 Validated scripts are maintained for project use.
3. Set up EDA environment.	3.1 <u>EDA tools and technology libraries</u> are configured according to project requirements. 3.2 Environment variables and library paths are verified before tool execution. 3.3 <u>Design projects</u> are created and organized within the EDA environment. 3.4 Tool configuration is tested to ensure proper operation. 3.5 EDA environment is prepared for RTL design and verification.
4. Write Verilog RTL modules.	4.1 <u>RTL requirements</u> are interpreted from the design specification.

	4.2 Verilog modules are developed using synthesizable coding practices. 4.3 <u>Combinational and sequential logic</u> are implemented according to design requirements. 4.4 RTL code is compiled and checked for syntax errors. 4.5 RTL modules are reviewed before simulation.
5. Design FSM and DFT-registers.	5.1 <u>FSM requirements</u> are interpreted from the design specification. 5.2 State machines are implemented using synthesizable RTL. 5.3 DFT registers and scan registers are developed according to design requirements. 5.4 FSM and DFT register functionality are verified through simulation. 5.5 <u>Design issues</u> are identified and corrected before verification.
6. Create and simulate RTL testbench.	6.1 <u>RTL testbench</u> is developed according to verification requirements. 6.2 Test stimuli and expected outputs are prepared for simulation. 6.3 <u>RTL simulation</u> is executed using the appropriate EDA tool. 6.4 Simulation waveforms and results are analyzed to verify functionality. 6.5 Functional issues are identified, corrected, and revalidated through simulation.

Range of Variables

Variable	Range May include but not limited to:
1. Linux/UNIX environment	1.1 Linux Workstation 1.2 UNIX Terminal 1.3 Shell Environment (Bash/Tcsh) 1.4 Remote Server 1.5 SSH Session 1.6 User Workspace
2. Project files	2.1 Design Files 2.2 Schematic Files 2.3 Layout Files 2.4 Technology Libraries 2.5 Simulation Files 2.6 Project Directories
3. Shell and TCL scripts	3.1 Shell Script (Bash/C Shell) 3.2 TCL Script 3.3 EDA Automation Script 3.4 Build Script 3.5 Verification Script

	3.6 Utility Script
4. Design and verification tasks	4.1 EDA Tool Execution 4.2 Simulation Automation 4.3 Synthesis Automation 4.4 Report Generation 4.5 File Management 4.6 Batch Processing
5. EDA tools and technology libraries	5.1 RTL Design Tools 5.2 Logic Simulation Tools 5.3 Synthesis Tools 5.4 Waveform Viewer 5.5 Standard Cell Libraries 5.6 Process Design Kit (PDK)
6. Design projects	6.1 Project Workspace 6.2 RTL Source Files 6.3 Verification Files 6.4 Script Directory 6.5 Library Directory 6.6 Project Database
7. RTL requirements	7.1 Design Specification 7.2 Functional Requirements 7.3 Interface Specification 7.4 Timing Requirements 7.5 Design Constraints 7.6 Module Architecture
8. Combinational and sequential logic	8.1 Multiplexer (MUX) 8.2 Flip-Flops 8.3 Registers 8.4 Finite State Machine (FSM)
9. FSM requirements	9.1 Design Specification 9.2 State Diagram 9.3 State Transition Table 9.4 Input/Output Requirements 9.5 Timing Requirements 9.6 Functional Requirements
10. Design issues	10.1 Functional Errors 10.2 State Transition Errors 10.3 RTL Coding Errors 10.4 Scan Connectivity Issues 10.5 Timing Issues 10.6 Corrective Actions
11. RTL testbench	11.1 Testbench Module 11.2 DUT (Design Under Test) 11.3 Clock Generator 11.4 Reset Generator 11.5 Test Tasks and Functions 11.6 Verification Environment

12. RTL simulation	12.1 RTL Simulator 12.2 Simulation Script 12.3 Simulation Run 12.4 Compilation and Elaboration 12.5 Simulation Log 12.6 Waveform Database
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Curricular Content Guide

1. Underpinning knowledge	1.1 Linux/UNIX fundamentals, command-line operations, and file management. 1.2 Shell and TCL scripting concepts for EDA automation. 1.3 EDA tool setup, technology libraries, and project environment configuration. 1.4 Verilog RTL coding principles for combinational and sequential logic. 1.5 FSM design, DFT registers, and scan register concepts. 1.6 RTL debugging, functional verification, and code validation techniques. 1.7 Project organization, coding standards, and design documentation practices.
2. Underpinning skills	2.1 Operating Linux/UNIX environments for EDA project development and file management 2.2 Configuring EDA tools, technology libraries, and project environments for RTL development 2.3 Writing synthesizable Verilog RTL modules for combinational and sequential logic 2.4 Compiling, debugging, and reviewing RTL code to ensure design quality 2.5 Creating RTL testbenches, test stimuli, and simulation environments for functional verification 2.6 Analyzing simulation waveforms to identify, debug, and correct functional issues 2.7 Applying RTL design, scripting, and verification techniques to support ASIC development and DFT implementation.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Tools & equipment 4.4 Job specifications and work instructions
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 operated Linux/UNIX environment. 1.2 applied shell and TCL scripting. 1.3 set up EDA environment. 1.4 write Verilog RTL modules. 1.5 designed FSM and DFT- registers. 1.6 created and simulated RTL testbench.
2. Methods of assessment	Methods of assessment should include: 2.1 Written Test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DEVELOP SYSTEMVERILOG TESTBENCH AND VERIFICATION ENVIRONMENT	Nominal Duration: 40 hrs.	Unit Code: SICIP-SCD-DVE-02-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to Develop SystemVerilog Testbench and Verification Environment. It specifically includes the tasks of applying SystemVerilog data types, applying interfaces and testbench structure, applying object-oriented programming, developing testbench architecture, executing simulations and analyze waveforms and debugging verification failures.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Apply SystemVerilog data types.	1.1 <u>SystemVerilog data types</u> are selected according to verification requirements. 1.2 Variables, arrays, structures, and enumerations are declared using appropriate data types. 1.3 Data types are applied to support testbench development. 1.4 <u>Source code</u> is compiled and verified for syntax correctness. 1.5 Data type implementation is reviewed before simulation.
2. Apply interfaces and testbench structure.	2.1 SystemVerilog interfaces are developed to connect design and testbench components. 2.2 <u>Testbench modules and components</u> are organized according to verification requirements. 2.3 Clock, reset, and communication signals are configured within the testbench. 2.4 Testbench structure is verified for connectivity and functionality. 2.5 <u>Testbench environment</u> is prepared for simulation.
3. Apply object-oriented programming.	3.1 Classes and objects are developed using SystemVerilog OOP concepts. 3.2 Inheritance, polymorphism, and encapsulation are applied in the verification environment. 3.3 <u>Class methods and constructors</u> are implemented according to verification requirements. 3.4 <u>OOP-based components</u> are integrated into the testbench. 3.5 OOP implementation is verified before simulation.
4. Develop testbench architecture.	4.1 <u>Verification components</u> are developed according to the testbench architecture. 4.2 Drivers, monitors, generators, and scoreboards are

	implemented as required. 4.3 Test scenarios and stimulus generation are configured according to verification objectives. 4.4 <u>Testbench architecture</u> is integrated and validated for functional verification. 4.5 Testbench environment is prepared for simulation execution.
5. Execute simulations and analyze waveforms.	5.1 <u>Simulation</u> is executed using the appropriate EDA tool. 5.2 Simulation waveforms and logs are reviewed to verify design behavior. 5.3 Functional results are analyzed against verification requirements. 5.4 <u>Verification reports</u> are generated and reviewed. 5.5 Simulation results are validated before debugging activities.
6. Debug verification failures.	6.1 <u>Verification failures</u> are identified from simulation results and reports. 6.2 Waveforms, logs, and debug information are analyzed to determine root causes. 6.3 <u>Testbench or RTL issues</u> are corrected according to verification requirements. 6.4 Simulation is repeated to verify corrective actions. 6.5 Verification results are documented and approved after successful debugging.

Range of Variables

Variable	Range
	May include but not limited to:
1. SystemVerilog data types	1.1 2-State Data Types (bit, int) 1.2 4-State Data Types (logic, reg) 1.3 Integer Data Types (byte, shortint, longint) 1.4 Real Data Types (real, shortreal) 1.5 String 1.6 Class Handle
2. Source code	2.1 SystemVerilog Compiler 2.2 Compilation Log 2.3 Syntax Checker 2.4 Error Messages 2.5 Warning Messages 2.6 Compilation Report
3. Testbench modules and components	3.1 Top-Level Testbench 3.2 Design Under Test (DUT) 3.3 Driver 3.4 Monitor 3.5 Generator 3.6 Scoreboard

4. Testbench environment	4.1 Simulation Environment 4.2 Verification Workspace 4.3 Compilation Scripts 4.4 Simulation Scripts 4.5 Waveform Configuration 4.6 Verification Configuration Files
5. Class methods and constructors	5.1 Constructor (new) 5.2 User-Defined Methods 5.3 Functions 5.4 Tasks 5.5 Static Methods 5.6 Virtual Methods
6. OOP-based components	6.1 Transaction Class 6.2 Generator 6.3 Driver 6.4 Monitor 6.5 Scoreboard 6.6 Environment Class
7. Verification components	7.1 Driver 7.2 Monitor 7.3 Generator 7.4 Scoreboard 7.5 Environment 7.6 Agent
8. Testbench architecture	8.1 Top-Level Testbench 8.2 Design Under Test (DUT) 8.3 Interface 8.4 Verification Components Integration 8.5 Communication Channels 8.6 Functional Verification Flow
9. Simulation	9.1 RTL Simulation 9.2 Functional Simulation 9.3 Batch Simulation 9.4 Interactive Simulation 9.5 Regression Simulation 9.6 EDA Simulation Tool
10. Verification reports	10.1 Simulation Report 10.2 Functional Verification Report 10.3 Coverage Report 10.4 Assertion Report 10.5 Test Summary Report 10.6 Verification Log
11. Verification failures	11.1 Assertion Failures 11.2 Functional Mismatches 11.3 Test Case Failures 11.4 Protocol Violations 11.5 Simulation Errors

	11.6 Coverage Gaps
12. Testbench or RTL issues	12.1 Testbench Errors 12.2 RTL Coding Errors 12.3 Interface Errors 12.4 Timing Issues 12.5 Reset/Clock Issues 12.6 Stimulus Generation Errors

Curricular Content Guide

1. Underpinning knowledge	1.1 SystemVerilog data types, variables, arrays, structures, and enumerations. 1.2 SystemVerilog interfaces and testbench architecture fundamentals. 1.3 Object-Oriented Programming (OOP) concepts. 1.4 Verification components such as drivers, monitors, generators, and scoreboards. 1.5 Waveform analysis, simulation logs, and verification report interpretation. 1.6 Debugging techniques for RTL and testbench verification failures. 1.7 Verification methodology, coding standards, and documentation practices.
2. Underpinning skills	2.1 Applying SystemVerilog data types, arrays, structures, and enumerations in verification environments 2.2 Developing SystemVerilog interfaces and structured testbench architectures 2.3 Applying object-oriented programming (OOP) concepts. 2.4 Designing verification components such as drivers, monitors, generators, and scoreboards 2.5 Creating and integrating functional verification testbench environments 2.6 Validating simulation results through repeated testing and functional verification 2.7 Applying SystemVerilog verification methodologies to ensure accurate and reliable ASIC design verification
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Specifications and work instructions 4.3 Learning materials 4.4 Tools and equipment
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 applied SystemVerilog data types. 1.2 applied interfaces and testbench structure. 1.3 applied object-oriented programming. 1.4 developed testbench architecture. 1.5 executed simulations and analyze waveforms. 1.6 debugged verification failures.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM COVERAGE-DRIVEN AND UVM VERIFICATION	Nominal Duration: 44 hrs.	Unit Code: SICIP-SCD-DVE-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to Perform Coverage-Driven and UVM Verification. It specifically includes the tasks of Write assertions, implementing functional coverage, applying coverage-driven verification, applying UVM methodology, developing UVM components and sequences and implementing checking and debugging.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Write assertions.	1.1 <u>Verification requirements</u> are analyzed before developing assertions. 1.2 SystemVerilog assertions are written according to design specifications. 1.3 Assertions are integrated into the <u>verification environment</u> . 1.4 Assertion results are analyzed during simulation. 1.5 Assertion violations are identified and documented.
2. Implement functional coverage.	2.1 <u>Functional coverage requirements</u> are identified from the verification plan. 2.2 Covergroups and coverpoints are developed according to verification objectives. 2.3 <u>Coverage models</u> are integrated into the testbench. 2.4 Functional coverage data is collected during simulation. 2.5 Coverage results are reviewed against verification targets.
3. Apply coverage-driven verification.	3.1 <u>Coverage goals</u> are established according to verification requirements. 3.2 Test scenarios are developed to improve functional coverage. 3.3 Coverage reports are analyzed to identify verification gaps. 3.4 Additional tests are executed to achieve coverage objectives. 3.5 <u>Coverage closure</u> is verified before completing verification.
4. Apply UVM methodology.	4.1 <u>UVM methodology</u> is applied according to verification requirements. 4.2 UVM base classes and verification phases are configured. 4.3 <u>UVM environment</u> is created using standard

	verification architecture. 4.4 UVM configuration and factory mechanisms are applied where required. 4.5 UVM environment is verified before simulation.
5. Develop UVM components and sequences.	5.1 <u>UVM components</u> are developed. 5.2 UVM sequences and sequence items are created according to the verification plan. 5.3 UVM components are integrated into the verification environment. 5.4 Simulation is executed to validate UVM functionality. 5.5 UVM implementation is reviewed for completeness and correctness.
6. Implement checking and debugging.	6.1 <u>Checkers and scoreboards</u> are implemented to verify design behavior. 6.2 Simulation logs, waveforms, and reports are analyzed to identify verification failures. 6.3 <u>Verification issues</u> are debugged and corrective actions are applied. 6.4 Simulation is repeated to verify issue resolution. 6.5 Final verification results are reviewed and documented.

Range of Variables

Variable	Range
	May include but not limited to:
1. Verification requirements	1.1 Functional Specification 1.2 Verification Plan 1.3 Design Requirements 1.4 Interface Specification 1.5 Protocol Requirements 1.6 Test Objectives
2. Verification environment	2.1 RTL Testbench 2.2 SystemVerilog Testbench 2.3 UVM Environment 2.4 Design Under Test (DUT) 2.5 Interface 2.6 Simulation Environment
3. Functional coverage requirements	3.1 Verification Plan 3.2 Verification Objectives 3.3 Functional Specification 3.4 Coverage Targets 3.5 Test Plan 3.6 Design Features
4. Coverage models	4.1 Functional Coverage Model 4.2 Testbench Components

	4.3 UVM Environment 4.4 Design Under Test (DUT) 4.5 Interface 4.6 Verification Environment
5. Coverage goals	5.1 Functional Coverage Targets 5.2 Code Coverage Targets 5.3 Assertion Coverage Targets 5.4 Verification Objectives 5.5 Test Plan 5.6 Coverage Metrics
6. Coverage closure	6.1 Coverage Closure Report 6.2 Coverage Metrics Review 6.3 Verification Signoff 6.4 Test Completion Report 6.5 Final Coverage Summary 6.6 Verification Approval Documentation
7. UVM methodology	7.1 UVM Standard Methodology 7.2 Verification Plan 7.3 UVM Testbench Architecture 7.4 Transaction-Based Verification 7.5 Reusable Verification Components 7.6 UVM Coding Guidelines
8. UVM environment	8.1 Test 8.2 Environment (Env) 8.3 Agent 8.4 Driver 8.5 Monitor 8.6 Scoreboard
9. UVM components	9.1 UVM Agent 9.2 UVM Driver 9.3 UVM Monitor 9.4 UVM Scoreboard 9.5 UVM Sequencer 9.6 UVM Environment
10. Checkers and scoreboards	10.1 Assertion Checkers 10.2 Protocol Checkers 10.3 Functional Checkers 10.4 Scoreboard 10.5 Reference Model 10.6 Result Comparator
11. Verification issues	11.1 Functional Errors 11.2 Assertion Failures 11.3 Protocol Violations 11.4 Testbench Errors 11.5 RTL Design Errors 11.6 Corrective Actions

Curricular Content Guide

1. Underpinning knowledge	1.1 SystemVerilog Assertions (SVA) principles and assertion-based verification. 1.2 Functional coverage concepts, covergroups, coverpoints, and coverage analysis. 1.3 Coverage-driven verification methodology and coverage closure techniques. 1.4 UVM fundamentals, architecture, phases, and factory mechanism. 1.5 Checkers, scoreboards, and functional verification techniques. 1.6 Simulation, waveform analysis, debugging, and verification report interpretation. 1.7 Verification best practices, coding standards, and documentation.
2. Underpinning skills	2.1 Writing and integrating SystemVerilog assertions for functional verification 2.2 Developing functional coverage models using covergroups and coverpoints 2.3 Applying Universal Verification Methodology (UVM) principles and verification architecture 2.4 Developing UVM components, sequences, and sequence items for reusable verification environments 2.5 Analyzing simulation logs, waveforms, assertions, and coverage reports to identify verification issues 2.6 Debugging verification failures and validating corrective actions through repeated simulation 2.7 Applying advanced SystemVerilog, UVM, and coverage techniques.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 Job specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 write assertions. 1.2 implemented functional coverage. 1.3 applied coverage-driven verification. 1.4 applied UVM methodology. 1.5 developed UVM components and sequences. 1.6 implemented checking and debugging.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY ADVANCED ASIC VERIFICATION TECHNIQUES	Nominal Duration: 24 hrs.	Unit Code: SICIP-SCD-DVE-04-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to Apply Advanced ASIC Verification Techniques. It specifically includes the tasks of interpreting timing and constraints, performing CDC/RDC verification, implementing low-power verification concepts and executing lint, formal and equivalence checking.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret timing and constraints.	1.1 <u>Timing requirements and design constraints</u> are interpreted from the design specification. 1.2 Clock definitions, timing paths, and timing exceptions are identified. 1.3 Setup time, hold time, and timing reports are analyzed. 1.4 <u>Constraint files</u> are reviewed for completeness and correctness. 1.5 Timing analysis results are verified against design requirements.
2. Perform CDC/RDC verification.	2.1 <u>Clock Domain Crossing (CDC) and Reset Domain Crossing (RDC) requirements</u> are identified. 2.2 CDC and RDC verification tools are configured according to project requirements. 2.3 CDC and RDC violations are identified through verification reports. 2.4 <u>Synchronization issues</u> are analyzed and corrected. 2.5 Verification results are reviewed to ensure domain crossing compliance.
3. Implement low-power verification concepts.	3.1 <u>Low-power verification requirements</u> are identified from the design specification. 3.2 Power intent and low-power constraints are applied according to project requirements. 3.3 Low-power verification checks are executed using the appropriate EDA tool. 3.4 <u>Verification results</u> are analyzed to identify power-related issues. 3.5 Low-power verification results are validated before signoff.
4. Execute lint, formal and equivalence checking.	4.1 <u>RTL linting</u> is performed to identify coding issues and rule violations. 4.2 Formal verification is executed according to verification requirements.

	4.3 Logical Equivalence Checking (LEC) is performed between RTL and synthesized designs. 4.4 Verification reports are analyzed and identified issues are corrected. 4.5 Lint, formal, and equivalence checking results are reviewed and documented.
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Range of Variables

Variable	Range May include but not limited to:
1. Timing requirements and design constraints	1.1 Design Specification 1.2 Timing Constraints 1.3 Clock Constraints 1.4 Input/Output Delay Constraints 1.5 Design Rules 1.6 SDC (Synopsys Design Constraints) File
2. Constraint files	2.1 SDC File 2.2 Clock Constraints 2.3 Input Delay Constraints 2.4 Output Delay Constraints 2.5 Timing Exception Constraints 2.6 Constraint Validation Report
3. Clock Domain Crossing (CDC) and Reset Domain Crossing (RDC) requirements	3.1 Clock Domain Crossing (CDC) 3.2 Reset Domain Crossing (RDC) 3.3 Clock Domains 3.4 Reset Domains 3.5 Synchronization Requirements 3.6 Verification Plan
4. Synchronization issues	4.1 Two-Flip-Flop Synchronizer 4.2 Reset Synchronizer 4.3 Handshake Synchronization 4.4 FIFO Synchronization 4.5 RTL Modifications 4.6 Constraint Updates
5. Low-power verification requirements	5.1 Design Specification 5.2 Power Specification 5.3 Verification Plan 5.4 Low-Power Design Requirements 5.5 Power Management Requirements 5.6 Signoff Criteria
6. Verification results	6.1 Power Verification Report 6.2 Power-Aware Simulation Results 6.3 Isolation Violations 6.4 Retention Violations 6.5 Power State Errors 6.6 Verification Logs

7. RTL linting	7.1 RTL Lint Tool 7.2 Coding Rule Checks 7.3 Syntax Checks 7.4 Coding Standard Compliance 7.5 Lint Violations 7.6 Lint Report
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Curricular Content Guide

1. Underpinning knowledge	1.1 Timing analysis concepts, including clocks, setup time, hold time, and timing constraints. 1.2 Clock Domain Crossing (CDC) and Reset Domain Crossing (RDC) verification principles. 1.3 RTL linting principles, coding standards, and rule checking. 1.4 Interpretation of timing, CDC/RDC, lint, formal, and verification reports. 1.5 Techniques for identifying and resolving timing and verification issues. 1.6 Verification sign-off requirements, quality assurance, and documentation practices.
2. Underpinning skills	2.1 Analyzing setup time, hold time, clock definitions, timing paths, and timing exceptions 2.2 Performing Clock Domain Crossing (CDC) and Reset Domain Crossing (RDC) verification 2.3 Applying low-power verification concepts, power intent, and low-power constraints 2.4 Executing RTL linting to identify coding issues and rule violations 2.5 Performing formal verification and Logical Equivalence Checking (LEC) for design validation 2.6 Analyzing lint, CDC/RDC, low-power, formal, and equivalence checking reports 2.7 Validating verification results to ensure design quality and signoff readiness
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Learning manual 4.3 Specifications and work instructions
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Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interpreted timing and constraints. 1.2 performed CDC/RDC verification. 1.3 implemented low-power verification concepts. 1.4 executed lint, formal and equivalence checking.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IMPLEMENT DESIGN FOR TESTABILITY (DFT)	Nominal Duration: 36 hrs.	Unit Code: SICIP-SCD-DVE-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to Implement Design for Testability (DFT). It specifically includes the tasks of explaining scan architecture, performing scan insertion setup, debugging scan DRC and violations, applying ATPG and fault model concepts, generating and analyze ATPG coverage, applying compression and timing-aware testing.		

Elements and Performance Criteria

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain scan architecture.	1.1 <u>Scan architecture and DFT requirements</u> are interpreted according to design specifications. 1.2 Scan cells, scan chains, and scan modes are identified. 1.3 Scan architecture is related to ASIC testing and fault detection. 1.4 <u>Scan design documentation</u> is reviewed before implementation. 1.5 Scan architecture is verified against project requirements.
2. Perform scan insertion setup.	2.1 <u>Design files and DFT constraints</u> are prepared for scan insertion. 2.2 Scan insertion is configured using the appropriate EDA tool. 2.3 <u>Scan chains and test signals</u> are generated according to design requirements. 2.4 Scan insertion reports are reviewed for correctness. 2.5 can implementation is verified before DFT validation.
3. Debug scan DRC and violations.	3.1 <u>Scan DRC</u> is executed using the appropriate DFT tool. 3.2 Scan DRC reports are analyzed to identify violations. 3.3 <u>Scan connectivity and design issues</u> are corrected according to DFT requirements. 3.4 Corrective actions are verified through re-execution of DRC checks. 3.5 Scan DRC results are documented and approved.
4. Apply ATPG and fault model concepts.	4.1 <u>Fault models</u> are identified according to testing requirements. 4.2 ATPG setup is configured using the appropriate DFT tool. 4.3 <u>Test patterns</u> are generated based on selected fault models. 4.4 ATPG reports are reviewed to verify test generation.

	4.5 ATPG implementation is validated against project requirements.
5. Generate and analyze ATPG coverage.	5.1 ATPG execution is performed to generate test patterns. 5.2 Fault coverage reports are generated and reviewed. 5.3 Coverage gaps and untested faults are identified and analyzed. 5.4 Test patterns are refined to improve fault coverage. 5.5 Final ATPG coverage results are verified and documented.
6. Apply compression and timing-aware testing.	6.1 Test compression techniques are applied according to DFT requirements. 6.2 Timing-aware test configurations are implemented using the appropriate DFT tool. 6.3 Compressed test patterns are generated and validated. 6.4 Timing-related test issues are identified and resolved. 6.5 Final test results are verified to ensure required test quality and coverage.

Range of Variables

Variable	Range
	May include but not limited to:
1. Scan architecture and DFT requirements	1.1 Design Specification 1.2 DFT Requirements 1.3 Scan Architecture 1.4 Testability Requirements 1.5 Verification Plan 1.6 Project Constraints
2. Scan design documentation	2.1 DFT Specification 2.2 Scan Chain Report 2.3 Scan Architecture Diagram 2.4 Design Documentation 2.5 Test Plan 2.6 Implementation Guidelines
3. Design files and DFT constraints	3.1 RTL Design Files 3.2 Technology Libraries 3.3 DFT Constraint Files 3.4 Scan Configuration Files 3.5 Timing Constraints 3.6 Design Netlist
4. Scan chains and test signals	4.1 Scan Chains 4.2 Scan Flip-Flops 4.3 Scan Enable Signal 4.4 Scan-In (SI) 4.5 Scan-Out (SO) 4.6 Test Clock

5. Scan DRC	5.1 DFT Rule Check (DRC) Tool 5.2 Scan Rule Checks 5.3 Design Rule Validation 5.4 Scan Configuration 5.5 DFT Constraints 5.6 Verification Setup
6. Scan connectivity and design issues	6.1 Broken Scan Chains 6.2 Unconnected Scan Cells 6.3 Scan Enable Issues 6.4 Test Signal Connectivity Errors 6.5 Clock and Reset Issues 6.6 DFT Constraint Violations
7. Fault models	7.1 Stuck-at Fault Model 7.2 Transition Fault Model 7.3 Bridging Fault Model 7.4 Path Delay Fault Model 7.5 Cell-Aware Fault Model 7.6 IDDQ Fault Model
8. Test patterns	8.1 Scan Test Patterns 8.2 Functional Test Patterns 8.3 Compressed Test Patterns 8.4 Deterministic Test Patterns 8.5 Random Test Patterns 8.6 Pattern Database
9. ATPG execution	9.1 ATPG Tool 9.2 Test Pattern Generation 9.3 Scan Pattern Generation 9.4 Fault Simulation 9.5 Batch Execution 9.6 ATPG Run Configuration
10. Coverage gaps and untested faults	10.1 Untested Faults 10.2 Redundant Faults 10.3 Unobservable Faults 10.4 Uncontrollable Faults 10.5 Coverage Gaps 10.6 Fault Classification Report
11. Test compression techniques	11.1 Scan Compression 11.2 Embedded Deterministic Test (EDT) 11.3 X-Masking 11.4 X-Bounding 11.5 Test Data Decompression 11.6 Output Response Compaction
12. Compressed test patterns	12.1 Compressed Scan Patterns 12.2 Transition Test Patterns 12.3 Stuck-at Test Patterns 12.4 Pattern Validation 12.5 Pattern Database

Curricular Content Guide

1. Underpinning knowledge	1.1 Scan architecture concepts, including scan cells, scan chains, and scan modes. 1.2 DFT principles, scan insertion flow, and DFT constraints. 1.3 Scan DRC, scan connectivity verification, and violation debugging techniques. 1.4 Fault coverage analysis and ATPG report interpretation. 1.5 Test compression and timing-aware testing concepts. 1.6 DFT verification, test quality, and coverage optimization techniques. 1.7 DFT documentation, validation, and sign-off requirements.
2. Underpinning skills	2.1 Configuring and performing scan insertion using DFT tools and design constraints 2.2 Analyzing and debugging scan DRC reports, connectivity issues, and design violations 2.3 Applying ATPG concepts, fault models, and test pattern generation techniques 2.4 Applying test compression and timing-aware testing methodologies for efficient test implementation 2.5 Interpreting DFT, scan insertion, ATPG, and coverage reports to validate test readiness 2.6 Applying industry-standard DFT and ATPG workflows to ensure ASIC testability and signoff readiness
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 explained scan architecture. 1.2 performed scan insertion setup. 1.3 debugged scan DRC and violations. 1.4 applied ATPG and fault model concepts. 1.5 generated and analyzed ATPG coverage. 1.6 applied compression and timing-aware testing.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY ADVANCED TEST AND SILICON VALIDATION TECHNIQUES	Nominal Duration: 16 hrs.	Unit Code: SICIP-SCD-DVE-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to Apply Advanced Test and Silicon Validation Techniques. It specifically includes the tasks of explaining boundary scan and JTAG, implementing MBIST and LBIST concepts, interpreting OPCG/OCC concepts and performing post-silicon diagnosis.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Explain boundary scan and JTAG.	1.1 <u>Boundary scan and JTAG concepts</u> are interpreted according to test requirements. 1.2 Boundary scan architecture and JTAG components are identified. 1.3 JTAG operations and test access mechanisms are explained. 1.4 <u>Boundary scan applications</u> are related to chip testing and debugging. 1.5 Boundary scan and JTAG documentation are interpreted correctly.
2. Implement MBIST and LBIST concepts.	2.1 <u>MBIST and LBIST concepts</u> are identified according to design requirements. 2.2 Memory and logic BIST architectures are interpreted before implementation. 2.3 MBIST and LBIST configurations are applied using the appropriate DFT tools. 2.4 <u>BIST reports</u> are reviewed to verify implementation results. 2.5 MBIST and LBIST implementation is validated according to project requirements.
3. Interpret OPCG/OCC concepts.	3.1 <u>OPCG and OCC concepts</u> are interpreted according to DFT requirements. 3.2 Clock generation and clock control mechanisms are identified. 3.3 OPCG/OCC configurations are related to scan testing and ATPG. 3.4 Timing and clock control reports are reviewed for correctness. 3.5 <u>OPCG/OCC implementation requirements</u> are verified before testing.

4. Perform post-silicon diagnosis.	4.1 <u>Post-silicon test results and failure reports</u> are reviewed.
	4.2 Silicon defects and failure causes are analyzed using diagnostic data.
	4.3 <u>Diagnostic tools</u> are used to identify and isolate faults.
	4.4 Corrective actions and design improvements are recommended based on diagnosis results.
	4.5 Post-silicon diagnosis results are documented according to project requirements.

Range of Variables

Variable	Range
	May include but not limited to:
1. Boundary scan and JTAG concepts	1.1 Boundary Scan 1.2 JTAG (IEEE 1149.1) 1.3 Test Access Port (TAP) 1.4 On-Chip Testing 1.5 Board-Level Testing 1.6 Debug Interface
2. Boundary scan applications	2.1 Board-Level Interconnect Testing 2.2 Manufacturing Testing 2.3 Device Debugging 2.4 In-System Programming (ISP) 2.5 Fault Isolation 2.6 System Validation
3. MBIST and LBIST concepts	3.1 Memory Built-In Self-Test (MBIST) 3.2 Logic Built-In Self-Test (LBIST) 3.3 Built-In Self-Test (BIST) 3.4 Self-Test Controller 3.5 Test Coverage 3.6 Fault Detection
4. BIST reports	4.1 MBIST Report 4.2 LBIST Report 4.3 Fault Coverage Report 4.4 Test Summary Report 4.5 Execution Log 4.6 Error Report
5. OPCG and OCC concepts	5.1 On-Product Clock Generator (OPCG) 5.2 On-Chip Clock Controller (OCC) 5.3 At-Speed Testing 5.4 Test Clock Generation 5.5 Clock Control Logic 5.6 DFT Requirements
6. OPCG/OCC implementation requirements	6.1 Clock Functionality Verification 6.2 Timing Compliance Verification 6.3 Scan Test Readiness

	6.4 ATPG Compatibility Check 6.5 DFT Signoff Report 6.6 Implementation Documentation
7. Post-silicon test results and failure reports	7.1 Silicon Test Results 7.2 Failure Analysis Report 7.3 Production Test Report 7.4 Yield Report 7.5 Diagnostic Report 7.6 Test Logs
8. Diagnostic tools	8.1 Failure Diagnosis Tool 8.2 ATPG Diagnosis Tool 8.3 Logic Analyzer 8.4 JTAG Debug Tool 8.5 Scan Diagnosis Tool 8.6 Silicon Debug Software

Curricular Content Guide

1. Underpinning knowledge	1.1 Fundamentals of Boundary Scan and JTAG technology. 1.2 Boundary Scan architecture and JTAG components. 1.3 JTAG operations and Test Access Port (TAP) controller functions. 1.4 Applications of Boundary Scan for chip testing, debugging, and board-level testing. 1.5 Interpretation of JTAG and Boundary Scan documentation and standards.
2. Underpinning skills	2.1 Applying MBIST and LBIST concepts for memory and logic built-in self-test implementation 2.2 Configuring and validating MBIST/LBIST using industry-standard DFT tools 2.3 Interpreting timing, clock control, and DFT reports to support test implementation 2.4 Performing post-silicon diagnosis using test results, failure reports, and diagnostic tools 2.5 Analyzing silicon defects and recommending corrective actions for design improvement 2.6 Applying advanced DFT, BIST, JTAG, and post-silicon validation techniques to ensure ASIC test quality and reliability
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace

	3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 explained boundary scan and JTAG. 1.2 implemented MBIST and LBIST concepts. 1.3 interpreted OPCG/OCC concepts. 1.4 performed post-silicon diagnosis.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: COMPLETE ASIC VERIFICATION AND DFT PROJECT	Nominal Duration: 20 hrs.	Unit Code: SICIP-SCD-DVE-07-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to Complete ASIC Verification and DFT Project. It specifically includes the tasks of planning project and preparing architecture, executing verification and DFT implementation, validating verification results, preparing project documentation and presenting project outcomes.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Plan project and prepare architecture.	1.1 <u>Project requirements and design specifications</u> are reviewed before implementation. 1.2 Verification objectives, DFT requirements, and project architecture are planned. 1.3 Verification environment, DFT flow, and required EDA tools are prepared. 1.4 Project milestones, timelines, and deliverables are defined according to project requirements. 1.5 <u>Project plan</u> is reviewed and approved before implementation.
2. Execute verification and DFT implementation.	2.1 <u>Verification environment and DFT implementation</u> are executed according to the project plan. 2.2 RTL verification, DFT insertion, ATPG, and simulation activities are performed using appropriate EDA tools. 2.3 <u>Verification and DFT issues</u> are identified and resolved during implementation. 2.4 Design and test reports are generated and reviewed throughout the project. 2.5 Implementation results are verified before project completion.
3. Validate verification results.	3.1 Simulation, coverage, and DFT reports are reviewed against project requirements. 3.2 Verification, fault coverage, and test quality are evaluated using project metrics. 3.3 <u>Outstanding verification and DFT issues</u> are identified and corrected. 3.4 Final verification and DFT results are validated before project closure. 3.5 <u>Project outputs</u> are confirmed to meet design and testing objectives.

4. Prepare project documentation.	4.1 Project reports, verification data, DFT results, and design files are compiled. 4.2 Project methodology, implementation activities, and outcomes are documented clearly. 4.3 <u>Technical documents</u> are reviewed for accuracy and completeness. 4.4 Project portfolio and supporting deliverables are prepared according to organizational standards. 4.5 Final project documentation is submitted according to assessment requirements.
5. Present project outcomes.	5.1 Project objectives, verification methodology, and DFT implementation are presented clearly. 5.2 Design decisions, verification results, and DFT outcomes are explained using technical evidence. 5.3 Technical questions and assessment feedback are addressed professionally. 5.4 Lessons learned and recommendations for future improvements are discussed. 5.5 <u>Final project presentation and design defense</u> are completed according to assessment requirements.

Range of Variables

Variable	Range May include but not limited to:
1. Project requirements and design specifications	1.1 Functional Requirements 1.2 Verification Requirements 1.3 DFT Requirements 1.4 Design Constraints 1.5 Technical Documentation
2. Project plan	2.1 Project Plan Document 2.2 Resource Allocation 2.3 Risk Assessment 2.4 Implementation Schedule 2.5 Approval Records 2.6 Project Management Documentation
3. Verification environment and DFT implementation	3.1 SystemVerilog/UVM Verification Environment 3.2 DFT Implementation Flow 3.3 Verification Testbench 3.4 Scan Insertion Environment 3.5 Simulation Environment 3.6 Project Configuration
4. Verification and DFT issues	4.1 RTL Functional Errors 4.2 Scan Chain Violations 4.3 ATPG Errors 4.4 Coverage Gaps 4.5 Assertion Failures

	4.6 Corrective Actions
5. Outstanding verification and DFT issues	5.1 Functional Failures 5.2 Coverage Gaps 5.3 DFT Rule Violations 5.4 ATPG Failures 5.5 Scan Chain Issues 5.6 Corrective Actions
6. Project outputs	6.1 Verified RTL Design 6.2 Verified Testbench 6.3 DFT-Implemented Design 6.4 Test Pattern Package 6.5 Final Technical Documentation 6.6 Project Deliverables
7. Technical documents	7.1 Technical Report 7.2 Design Documentation 7.3 Verification Documentation 7.4 DFT Documentation 7.5 Review Checklist 7.6 Project Records
8. Final project presentation and design defense	8.1 Presentation Slides 8.2 Project Portfolio 8.3 Technical Documentation 8.4 Demonstration Materials 8.5 Assessment Presentation 8.6 Design Defense Records

Curricular Content Guide

1. Underpinning knowledge	1.1 Verification project planning and design specification analysis. 1.2 DFT concepts, verification objectives, and project architecture. 1.3 Verification environment, DFT flow, and EDA tool preparation. 1.4 Project planning, milestones, timelines, and deliverables. 1.5 Project documentation, review, and approval procedures.
2. Underpinning skills	2.1 Planning ASIC verification and DFT projects based on design specifications and project objectives 2.2 Preparing verification environments, DFT flows, EDA tools, and project architectures 2.3 Executing RTL verification, DFT insertion, ATPG, and simulation using industry-standard EDA tools 2.4 Analyzing simulation, coverage, fault coverage, and DFT reports to validate design quality 2.5 Evaluating verification metrics and confirming

	compliance with design and testing objectives 2.6 Preparing technical documentation, verification reports, DFT results, and project deliverables 2.7 Demonstrating project management, technical communication, documentation, and problem-solving skills.
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 planned project and prepared architecture. 1.2 executed verification and DFT implementation. 1.3 validated verification results. 1.4 prepared project documentation. 1.5 presented project outcomes.
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	DFT and Verification Engineering Program for ASIC Designers
Number of Trainees:	25

Course-wise Training Space:

- Computer Lab : 400 sft
- Workshop : 700-800 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Computer/Laptops	01
2.	Printer	01
3.	Multimedia projector	01
4.	High-performance Desktop/Laptop	25
5.	Linux Operating System (Ubuntu/RHEL/Rocky Linux)	25
6.	ASIC Verification & DFT EDA Software Suite (Simulation, UVM, Lint, CDC/RDC, Formal Verification, DFT, ATPG, Waveform Debug)	25
7.	Verilog/SystemVerilog Simulation Tool	25
8.	UVM Verification Environment	25
9.	CDC/RDC Verification Tool	25
10.	Formal Verification & Equivalence Checking Tool	25
11.	DFT Scan Insertion Tool	25
12.	ATPG and Fault Simulation Tool	25
13.	Enterprise Router	01
14.	Waveform Debug and Analysis Tool	25
15.	Gigabit Network Switch (48-Port)	01
16.	Standard Cell Libraries and Verification IP (VIP) Repository	01
17.	Network Laser Printer	01
18.	Laser Pointer	01

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on DFT and Verification Engineering Program for ASIC Designers, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.